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Risk contagion in the banking network: New evidence from China

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ABSTRACT

Based on data from 111 Chinese banks over the 2013–2016 period, this paper estimates the interbank bilateral lending matrix using the maximum entropy method. The estimated matrix is used to simulate the effects of credit and liquidity shocks on China's banking network. Simulation results show that, under the extreme pressure scenario, the contagion arising from a liquidity shock is significantly stronger than the effect of a credit shock, indicating the importance of liquidity in the banking system. The contagion effect arising from a credit shock does not vary much over the sample period. However, the contagion effect arising from a liquidity shock decreases significantly, which could be attributed to contraction in interbank business due to stricter interbank business supervision. The simulation results also identify the most important and most vulnerable nodes of the banking system. An increase in the level of capital level can enhance the ability of banks to withstand credit and liquidity shocks. Our analysis also suggests that risk contagion faced by China's banks varies across banking network structures.

1. Introduction

China's banking industry has developed rapidly since the reform and opening up of the economy in 1978. By the end of 2018, there were 4,588 financial institutions in the banking industry, with a total asset size of US$39.08 trillion (CBIRC, 2019). The ratio of the assets of the Big-Four state-owned Chinese banks[1] to total assets of all banking institutions (CR4) decreased from 41.42% in 2013 to 35.34% in 2018, indicating an increase in the concentration of China's banking industry has decreased over time with increase in the level of competition.

Over the past few decades, China's banking industry has faced several challenges. The challenges include the collapse of China's Hainan Development Bank in 1998. Hainan Development Bank failure was the first major setback experienced by China's banking sector in the post-reform period. Non-performing loans are another challenge that contributed to real state bubble in China in 1990s (Allen et al., 2015, 2017). In the past decade or so, China's banking industry has been rocked by several scandals. For example, Bank of Qilu loan fraud in 2010, the Liuzhou bank loan fraud in 2014, the Agricultural Bank of China bills case in 2016, the illegal guarantee

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[1] The Big Four Chinese state-owned banks include the Industrial and Commercial Bank of China (ICBC), the China Construction Bank (CCB), the Agricultural Bank of China (ABC) and the Bank of China (BOC).

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case of China Guangfa bank and the illegal loan case of Shanghai Pudong Development bank in 2017. More recently, Hengfeng bank, Baoshang bank and Jinzhou bank have been involved in management-related crises. These scandals highlight the fact that China's banking sector is subject to significant risks. Accordingly, there is a need for continued research in the area of banking sector risk in China.

Banking sector growth in China has also resulted in expansion of the interbank market, which includes interbank lending and interbank business markets. Banks use the interbank lending market to carry out liquidity management (short term fund positions), and the interbank business market helps facilitate the settlements. The interbank lending market in China started functioning in 1994. Over the last 10 years or so, the interbank lending market has become the largest financial market in China. In 2018, the trading volume of the Chinese interbank market reached US$183.93 trillion, which includes the interbank lending transaction in the amount of US$20.29 trillion.[2] The interbank business market in China has expanded significantly since 2008. In 2012, the proportion of interbank assets to total assets of financial institutions in the banking industry was 15.13%, and the proportion of interbank liabilities to total liabilities was 13.87%. The proportion of some banks is much higher than the average of the banking industry. For example, the proportion of interbank assets and interbank liabilities of Industrial Bank, known as the "king of interbank", was 34.66% and 37.16%, respectively (PBC, 2013). The Chinese commercial banks have focused on the growth of interbank market as this market is subject to relatively less government regulations and hence the commercial banks are able to take advantage of regulatory arbitrage. The interbank business is subject to lower risk and therefore the bank capital utilization rate can be improved through the interbank business, which is also not constrained by either the deposit ratio or deposit reserves. Thus, interbank liabilities of Chinese commercial banks, especially those of small and medium-sized banks, have expanded rapidly in recent years.

Amstad & He (2019) provide a good analysis of the development of China's bond market (including the interbank and currency exchange market) and its inherent relationship with the banking system. Through a strong balance sheet connection among the banks, the fast growth of the interbank market has contributed to increased complexity of China's entire banking system. Owing to the strong balance sheet connection, any serious problem faced by an individual bank may quickly spread to other banks in the banking system. In other words, the fast growth of the interbank market has also increased financial risk faced by China's banking sector. For example, on 20 June 2013, tight liquidity of China's interbank market resulted in contagion. The maximum interbank overnight repo rate and 7-day repo rate, respectively, reached an unprecedented 30% and 28% The overnight and 7-day rates of Shanghai interbank offered rate (Shibor) rose to 13.44% and 11%, respectively. In the aftermath of this contagion, the banking sector regulators developed polices that resulted in increased supervision of the interbank business in 2014.[3]

In this paper, we first analyse the risk contagion effect of China's banking sector in 2013 from the perspective of banking network. We then investigate the risk contagion effect in China's banking industry in the context of the introduction of interbank business regulatory policies. Such an analysis allows one to evaluate the effectiveness of China's existing systemic risk prevention policies.

In the past decade or so, some existing studies have focused on risk contagion problems in the interbank markets of developed countries. For example, Furfine (2003) consider the case of the US, Boss et al. (2004) and Elsinger et al. (2006a) consider the case of Austria. Wells (2004) and Elsinger et al. (2006b) focus on the UK, Amundsen & Arnt (2005) focus on Denmark, Degryse & Nguyen (2007) focus on Belgium. Sheldon & Maurer (1998) and Müller (2006) consider the interbank market in Switzerland, Van Lelyveld & Liedorp (2006) consider the interbank market in Holland, Mistrulli (2011) consider the interbank market in Italy. Blavarg & Nimander (2002) and Frisell et al. (2007) analysed the risk contagion problem faced by the Swedish banking industry. Upper & Worms (2004) and Memmel & Stein (2008) focus on Germany and Toivanen (2009) considers the case of Finland. Paltalidis et al. (2015) evaluated the banking system of the euro zone.

While several studies have considered the case of developed countries, only a few existing studies focus on risk contagion problem in the interbank markets of developing countries. These studies include Guerrero-Gómez & Lopez-Gallo (2004), who consider the case of Mexico. Lublóy (2005) and Hausenblas et al. (2015), respectively, consider the case of Hungry and the Czech Republic. Moreover, existing studies on contagion effect can be categorized into two groups based on the degree of contagion effect. The first group, including Upper & Worms (2004) and Van Lelyveld & Liedorp (2006), report a serious contagion effect. Using a simulation model, they examine the effect of a shock to the German and Dutch banking systems. They find that the shock results in a loss of around three-quarters of the assets of the relevant banks. Degryse & Nguyen (2007), Mistrulli (2011) and Wells (2004) simulate the risk contagion problem in the interbank markets of Belgium, Italy, and the UK. They find that bankruptcy can lead to asset losses of 20%, 16% and 16%, respectively, under the worst-case scenario. The second group of studies such as Blavarg & Nimander (2002), Lublóy (2005), Sheldon & Maurer (1998), Amundsen & Arnt (2005) and Hausenblas et al. (2015) do not report a statistically significant contagion effect on the banking systems.

Differences between these two strands of literature can be mainly attributed to, among other things, the differences in the structure of banking systems and different estimation techniques used. The validity and reliability of simulation results are dependent on how interbank connections are measured and estimated.

In general, data on bilateral exposure between banks is confidential business information as the regulators do not require commercial banks to disclose this kind of information to public. Due to the unavailability of real data on bilateral exposure between banks, most existing studies tend to use the maximum entropy method (Jaynes, 1957) to approximate the data of bilateral exposure between banks. The data generated is then used to simulate the contagion effect of bank failures (Upper, 2011). The maximum entropy method (MEM) assumes that in the interbank market, the lending position of all individual banks within the network is evenly distributed,

[2] See https://www.ceicdata.com/en/china/national-interbank-funding-centre-nibfc-interbank-bond-turnover-daily.
[3] See CRBC (2014) and PBC (2014).

which allows one to construct the interbank bilateral lending matrix (i.e., a complete network is identified). The interbank market may be different from the real lending matrix due to relationship lending and other factors. Some studies have questioned the reliability of the MEM approach. For example, using data from the Italian interbank market, Mistrulli (2011) finds that MEM overestimates the risk contagion compared to the actual bilateral exposure data. However, while examining the risk contagion effect for four major banks from Europe (Germany, France, Italy and Spain), Paltalidis et al. (2015) found that the actual bilateral exposure data were not very different from MEM generated data and hence it is appropriate to use bilateral lending matrix based on the MEM approach.

The banking regulators in some countries require banks to report the interbank risk exposure data and hence a few available studies have used the real bilateral bank loans data for risk contagion analysis. These studies include Blavarg & Nimander (2002) and Frisell et al. (2007). Blavarg & Nimander and Frisell et al. analysed the risk contagion effect in the interbank market using regulatory reports on 15 largest exposures involving four major Swedish banks. Mistrulli (2011) analysed risk contagion in interbank market using the bilateral transaction data from Italy. Memmel & Stein (2008) analysed risk contagion in the German interbank market. Müller (2006) constructed a relatively complete interbank exposure matrix based on the data and credit lines of Swiss interbank loans. Müller also investigated the contagion channels of the Swiss banking system. Hausenblas et al. (2015) used data on bilateral lending between banks to examine the risk contagion in the banking sector of the Czech Republic. Van Lelyveld & Liedorp (2006) argue that the monitoring large exposure data can provide a good approximation of the full exposure data for the Dutch banking system. At present, financial regulatory authority in Italy, Hungary and Mexico require the banks to report complete bilateral exposure data. Of course, the use of higher quality data in simulations helps to provide a more accurate estimate of the contagion effects of systemic risk.

Some recent studies analyse and evaluate the methods used for constructing the interbank network. For example, Anand et al. (2018) used the network data on 25 financial markets to (i) consider the ability to reconstruct the network connections and exposure structure and (ii) rank different network reconstruction methods. Some studies propose new methods to construct the banking networks and analyse the risk contagion. Gorpe et al. (2019) propose a contagion mapping (CoMap) approach to describe the large exposure network involving the euro-area banks. This allows them to investigate the contagion possibility from external credit or financing shocks. Covi, Montagna et al. (2019), Covi, Gorpe et al. (2019) propose an adaptive contagion mapping method (A-CoMap). They examine how the precautionary behaviours of the banks may amplify or mitigate the contagion in euro-area banking network.

Maintaining the stability of the banking industry is of great significance to China's economic growth. However, so far, only a few studies have focused on risk contagion problem in China's interbank market. Li and Li (2005) and Ma et al. (2007) used the maximum entropy approach to estimate the bilateral lending matrix involving the Chinese banks. Li & Li found that the risk contagion probability of China's banking system is very low. Ma et al. (2007) argue that only the collapse of China Construction Bank and Bank of China can cause a large-scale contagion effect. Based on lending data from 19 banks in China and employing the threshold method to construct the interbank lending network, Li et al. (2011) compared the varying effects of random and selective shocks on stability of the banking network. They find the banking network is more stable under random shocks but less stable under selective shocks. Zou et al. (2019) used MEM to estimate the interbank loan data. They then combined the data with the cross-shareholding data of financial institutions to construct a double-layer network structure of China's financial system. Numerical simulation results showed that contagious losses in the double-layer network are higher than the sum of the losses infecting the interbank lending and cross-shareholding networks. Most of the excess losses were caused by the interbank lending channel. Sun (2020) investigates the contagion effects and systemic risk in China's commercial banks system based on the balance sheet data. The simulation results show that the contagious risk arising from an assumed bank failure is trivial, whereas the amplification effects of the losses due to the financial inter-linkage are significant.

Using the interbank assets, interbank liabilities and capital data collected from bank statements, and employing the MEM to estimate the bilateral lending matrix of China's interbank market, this paper examines the risk contagion effect of China's banking network arising from credit and liquidity shocks.

This paper makes four important contributions to the existing literature. First, studies on risk contagion in China's financial market are rare.[4] To the best of our knowledge, this is the first paper that simulates the risk contagion effect in China's banking system using the new measurement of banking capital over the 2013–2016 period. China's interbank business has developed rapidly since 2008, there was a liquidity crunch in the interbank market in June 2013, which led to the issuance of policies by the banking regulator in 2014 to strengthen the supervision of interbank business, which slowed down the development of interbank business. Another reason for using 2013 as a benchmark is that, from 2013, a new method of measuring banking sector capital has been used in China and hence our results are based on a consistent measure. Thus, this paper regards 2013 as a benchmark year to simulate the contagion effect of the banking sector risk.

Second, based on the classification of banks by China's banking regulatory authorities, we construct a unique dataset involving 111 banks. Our sample covers all ownership types: (i) state-owned large commercial banks; (ii) state-owned large policy banks such as the development bank and the postal savings bank; (iii) jointly-owned commercial banks; (iv) urban commercial banks; (v) rural commercial banks and (vi) foreign banks. The sample, which consists of 111 banks, is selected based on the importance of the banks. The total asset size of the banks included in our sample accounts for nearly 80% of the total assets of China's banking industry. Thus, the sample used in this paper can be regarded as highly representative. In contrast, Li and Li (2005) is based on a sample that includes only 17 banks. Ma et al. (2007) use a sample of 130 banks. The Chinese banking system lacks transparency and the situation in 2003 was even worse. As a result, the relevant data was not easily available.

Third, this paper examines the risk contagion effect of different network structures. MEM is used to estimate the complete network

[4] Using data from 2011 to 2013, Fei et al. (2015) compare the contagion risk in the interbank markets of China and the US. They find that shocks to Chinese banks with low capital adequacy ratio are likely to result in serious contagion.

structure. The threshold is set based on the size of bilateral lending between the banks. We construct a core-periphery index of the banking network, which allows us to identify two types of incomplete network structures. Simulation results based on these networks are then compared. The risk contagion effect is used to identify the relationship between the network structure and risk contagion. The existing studies do not explicitly consider the impact of network structure on risk contagion, but this issue has been highlighted by, among others, Li and Li (2005), Ma et al. (2007) and Sun (2020). Moreover, we use simulations to compare the net-capitals and the core tier 1 net capitals. This allows us to identify the contagion effect across capital levels.[5]

Last but not least, this paper compares the contagion effects arising from the credit and liquidity shock. We find that the contagion effect caused by liquidity shock is significantly stronger than the effect of the credit shock. Existing studies, e.g., Li and Li (2005) and Ma et al. (2007) simulate the contagion effect of the credit shock only. Furthermore, we find that some important and vulnerable nodes of the system change at different points of time. The simulations results presented in this paper are likely to be of significant interest to China's banking sector regulators tasked with monitoring the banking network and taking steps to prevent the emerging financial risks.

The remainder of this paper is organised as follows. Section 2 describes the data on interbank debt, credit and capital. The process involved in the construction of the interbank bilateral lending matrix of the Chinese banks is explained and the banking network constructed is analysed in Section 3. The constructed bilateral lending matrix is used for simulation of the risk contagion effect faced by of China's banking network under different network structures in Section 4. Section 5 contains conclusion and policy implications.

2. Data description

2.1. The sample

The China Banking Regulatory Commission (CBRC, 2020) data shows that, from 2013 onwards, a large number of financial institutions have been operating in China. These financial institutions include (i) 1 state development bank, China Development Bank (CDB); (ii) 2 policy banks, the Export-Import Bank of China (EIBC) and Agriculture Development Bank of China (ADBC); (iii) 5 large commercial banks; (iv) 1 postal savings bank, China Postal Savings Bank (PSB); (v) 12 joint-stock commercial banks;[6] (vi) 134 urban commercial banks; (vii) 1,114 rural commercial banks; (viii) 8 private banks; (ix) one Sino-German housing savings bank; (x) 40 rural cooperative banks; and (xi) 1,443 village banks. In addition, a large number of rural credit cooperatives, financial asset management companies, foreign-capital financial institutions, trust companies, enterprise group financial companies, financial leasing companies, money brokerage company, auto financing companies, consumer finance companies, loan companies and rural mutual funds cooperatives operate in China.

China's financial institutions include not only various types of financial institutions that offer banking services, but also a large number of small-scaled financial institutions that do not offer banking services. These financial institutions tend to focus on regional-oriented business. In this paper, we only consider banking-sector financial institutions with a national network and exclude non-banking financial institutions with regional network from our sample as these financial institutions might not have national contagion effect. Moreover, the two policy banks (EIBC and ADBC) are not included into our sample because of very large differences between policy banks and commercial banks in terms of the nature of operations, business areas, and regulatory requirements (but more importantly, the required data on these banks are unavailable). Similarly, the CDB has very large assets as well as interbank credit and debt. CDB has a nation-wide network and influence. Transformation of the CDB into a commercial bank started in 2008 and hence it is included into our sample.

Moreover, the size of bank assets is an important screening factor for our sample. Banks with large assets tend to have higher interbank creditor's rights and debts. And such banks tend to be closely linked with each other. Considering the relatively large scale of China's commercial banks, we use a threshold of RMB100 billion (approximately US$14.42 billion) in assets at the end of 2016 for inclusion in our sample. Large commercial banks and joint-stock commercial banks have asset in excess of US$14.42 billion. The assets of these banks, respectively, account for 37.29% and 18.72% of the financial assets of the banking sector in 2016. Thus, both banks are included in our sample. There are many city and rural-based commercial banks in China but most of these banks have relatively small assets and hence only sufficiently large city and rural commercial banks have been included in our sample. Another practical factor that we considered is the availability of annual financial reports for the 2013–2016 period. For our analysis, we require interbank creditor's rights and debts data as well as the bank capital data, which are usually published in the bank's annual financial reports. Banks that do not publish these data or have incomplete data are not included in our sample. Based on the considerations outlined above, our sample consists of 111 commercial banks, as shown in Table 1, which includes 5 state-owned large commercial banks, 1 development bank, 1 postal savings bank, 12 joint-stock commercial banks, 67 city commercial banks, 19 rural commercial banks, and 6 foreign banks.

[5] The literature on risk contagion effect is vast and covers other areas such as the stock market and currency exchange rates. For example, Lin, et al. (2019) investigate the direction of risk contagion among the Brent crude oil market, London gold market, and Chinese and European stock markets. Soylu and Güloğlu (2019) examine risk contagion in financial markets. Zorgati et al. (2019) examine the risk contagion effect in American and Asian stock markets, whereas Mim and Ben (2019) consider the case of the American and European markets. Zhang et al. (2020) consider the issue of volatility spillovers among G20 stock markets. This paper, however, focuses only on banking sector risk contagion.

[6] A public listed company requires the shareholders to be liable for the bank's debt.

Table 1
Bank Types and Names.

Type	Name
Large commercial banks (5)	Industrial and Commercial Bank of China (ICBC), China Construction Bank (CCB), Bank of China (BOC), Agricultural Bank of China (ABC), Bank of Communications (BCM)
Development bank (1)	China Development Bank (CDB)
Postal savings bank (1)	China Postal Savings Bank (PSB)
Joint-stock commercial banks (12)	Industrial Bank (CIB), China Merchants Bank (CMB), China CITIC Bank (CITIC), China Minsheng Bank China (MSB), Shanghai Pudong Development Bank (PDB), China Everbright Bank (CEB), Ping An Bank (PAB), Huaxia Bank (HXB), Guangfa Bank (GFB), Zhejiang Commercial Bank (ZCB), Hengfeng Bank (HFB), Bohai Bank (BHB)
City commercial banks (67)	Bank of Beijing(BOB), Shanghai Bank(SHB), Jiangsu Bank(JSB), Nanjing Bank(NJB), Shengjing Bank(SJB), Ningbo Bank (NBB), Huishang Bank(HSB), Hangzhou Bank(HZB), Tianjin Bank(TJB), Xiamen International Bank(XIB), Jinzhou Bank (JZB), Harbin Bank(HAB), Guangzhou Bank(GZB), Jilin Bank(JLB), Baoshang Bank(BSB), Changsha Bank(CSB), Chongqing Bank(CQB), Guiyang Bank(GYB), Zhengzhou Bank(ZZB), Chengdu Bank(CDB), Jiangxi Bank(JXB), Hebei Bank(HEB), Dalian Bank(DLB), Kunlun Bank(KLB), Qingdao Bank(QDB), Suzhou Bank(SZB), Huarong Xiangjiang Bank (XJB), Lanzhou Bank(LZB), Gansu Bank(GSB), Longjiang Bank(LJB), Dongguan Bank(DGB), Jiujiang Bank(JJB), Xi'an Bank(XAB), Hankou Bank(HKB), Qilu Bank(QLB), Sichuan Tianfu Bank(TFB), Tangshan Bank(TSB), Luoyang Bank(LYB), Wenzhou Bank(WZB), Fudian Bank(FDB), Guilin Bank(GLB), Xiamen Bank(XMB), Hubei Bank(HUB), Weihai City Commercial Bank(WHB), Chongqing Three Gorges Bank(TGB), Chang'an Bank(CAB), Jinshang Bank(JSB), Zhejiang Chouzhou Commercial Bank(ZCB), Taizhou Bank(TZB), Fuxin Bank(FXB), Handan Bank(HDB), Zhuhai Huarun Bank (HRB), Ningxia Bank(NXB), Guangxi Beibu Gulf Bank(BGB), Yingkou Bank(YKB), Zhejiang MinTai Commercial Bank (MTB), Guangdong Huaxing Bank(HXB), Zhejiang Tailong Commercial Bank(TLB), Inner Mongolia Bank(IMB), Cangzhou Bank(CZB), Liuzhou Bank(LIB), Ganzhou Bank(GAB), Rizhao Bank(RZB), Shaoxing Bank(SXB), Qishang Bank (QSB), Weifang Bank(WFB), Leshan City Commercial Bank(LSB)
Rural commercial banks (19)	Chongqing Rural Commercial Bank(CQR), Beijing Rural Commercial Bank(BJR), Shanghai Rural Commercial Bank (SHR), Chengdu Rural Commercial Bank(CDR), Guangzhou Rural Commercial Bank(GZR), Tianjin Rural Commercial Bank(TJR), Jiangsu Jiangnan Rural Commercial Bank(JNR), Guangdong Shunde Rural Commercial Bank(SDR), Hangzhou United Rural Commercial Bank(HZR), Guangdong Nanhai Rural Commercial Bank(NHR), Tianjin Binhai Rural Commercial Bank(BHR), Jiangsu Zijin Rural Commercial Bank (ZJR), Ningbo Yinzhou Rural Commercial Bank (YZR), Changshu Rural Commercial Bank(CSR), Wuxi Rural Commercial Bank (WXR), Xiamen Rural Commercial Bank (XMR), Shaoxing Ruifeng Rural Commercial Bank(RFR), Jiangyin Rural Commercial Bank (JYR), Jinan Rural Commercial Bank (JNR)
Foreign banks (6)	HSBC China (HSBC), Bank of East Asia China (BEA), Standard Chartered Bank China (SCB), CITIBank China (CITI), Nanyang Commercial Bank China (NCB), Sumitomo Mitsui Bank China (SMB)

2.2. Interbank credit, debt and bank capital data

Based on the provisions of CBRC, the interbank business in China is categorized into two groups: (i) the interbank financing business, which deals with interbank deposits, interbank lending, interbank borrowing, interbank refinance and reverse repurchases and (ii) the interbank investment business. Within the context of the balance sheets of commercial banks, the interbank creditor's rights include dues from the banks and other financial institutions, placements with banks and other financial institutions, and financial assets purchased under resale agreements. Interbank debts include dues to the banks and other financial institutions, placements from banks and other financial institutions, and financial assets sold under repurchase agreements.

In 2013, the Chinese banks started to issue interbank certificates of deposit for interbank financing. The issuance of interbank certificates of deposit by some banks grew rapidly. From an accounting perspective, the interbank certificates of deposit are viewed as bonds payable, but bonds payable include not only interbank certificates of deposit but also other bonds. Other bonds are issued to non-bank entities. As the interbank certificates of deposit are not listed separately in financial statements of banks and this data is merged into "bonds payable" it is not possible to include interbank certificates of deposit into interbank debt in our paper. Consequently, the interbank debt used in our paper is underestimated. The balance sheet data for our sample indicates that some banks have large number of bonds payable. For example, Bank of China, which is one of the 5 state-owned large commercial banks, has bonds payable exceeding the total amount of placements from banks and other financial institutions and financial assets sold under repurchase agreements.

Out of the 12 joint-stock commercial banks, only the China Merchants Bank, Guangfa Bank and Zhejiang Commercial Bank have the aggregate value of the bonds payable less than the sum of placements from banks and other financial institutions and financial assets sold under repurchase agreements. The value of the bonds payable by other banks is more than the sum of placements from banks and other financial institutions and financial assets sold under repurchase agreements. The aggregate value of the bonds payable of the Hengfeng Bank exceeds even the sum of its interbank debts.

Out of the 67 city commercial banks included in our sample, the aggregate values of the bonds payable by most of the banks exceed the sum of the placements from banks and other financial institutions and financial assets sold under repurchase agreements, and in some cases this value even exceed the sum of their interbank debts. The aggregate value of the bonds payable by the 19 rural commercial banks included in our sample are more than the sum of the placements from banks and other financial institutions and financial assets sold under repurchase agreements. The aggregate value of the bonds payable by the foreign banks is relatively small.

Prior to 2012, China's commercial banks use to calculate their net capital and net core capital based on the governing regulations of the Capital Adequacy Ratios for Commercial Banks. Since 2013, China's commercial banks are calculating their net capital, net tier 1

capital and net core tier 1 capital in accordance with the CBRC's *Capital Rules for Commercial Banks (Provisional)*. The total capital of commercial banks includes Core Tier 1 capital, Additional Tier 1 and Tier 2 capital. Core Tier 1 capital includes paid-in capital or common shares, capital reserves, surplus reserves, general risk reserves, undistributed profits, and the relevant part of the capital of minority shareholders. Additional Tier 1 capital includes additional Tier 1 capital instruments and their premiums, and the relevant part of the capital of minority shareholders. Tier 1 capital is the sum of the Core Tier 1 and Additional Tier 1 capital. Tier 2 capital includes Tier 2 capital instruments and their premiums, excess loan loss reserves, and the relevant part of the capital of minority shareholders. The net core tier 1 capital equals the core tier 1 capital minus capital deductions, which include the goodwill, other intangible assets (except land use rights), deferred tax assets relating to operating losses, and shortfall of loan loss provisions. Net core tier 1 capital plus additional tier 1 capital equals net tier 1 capital, net tier 1 capital plus tier 2 capital equals net capital. Net capital measures the ability of banks to withstand risks. Net core Tier 1 capital has the strongest ability to withstand risks. This follows from the fact that the net Tier 1 capital of many banks is very close to or even equivalent to the net core Tier 1 capital. In our simulation exercise later in the paper, capital data is compared only with net capital and net core Tier 1 capital.

3. Methodology

3.1. Maximum entropy method (MEM)

The Chinese banks do not release the interbank bilateral lending data, and the regulatory authorities do not have mandatory information disclosure requirements. However, a large number of banks do issue financial statements that include data on bank assets, liabilities, interbank assets and interbank liabilities. We follow the existing literature and use the MEM method to estimate China's interbank bilateral lending matrix.

The interbank bilateral lending relationship, without correction, is represented by matrix X as follows:

$$\begin{array}{c|ccccc|c} & & & \sum_i & & & \\ & x_{11} & \cdots & x_{1j} & \cdots & x_{1N} & a_1 \\ & \vdots & \ddots & \vdots & \cdots & \vdots & \vdots \\ X = & x_{i1} & x_{ij} & x_{iN} & & & a_i \\ & \vdots & \cdots & \ddots & \vdots & & \vdots \\ & x_{N1} & \cdots & x_{Nj} & \cdots & x_{NN} & a_N \\ \sum_j & l_1 & \cdots & l_j & \cdots & l_N & \end{array}$$

where x_{ij} represents the loans of bank i to bank j; N is the number of banks; $l_j = \sum_i x_{ij}$ and $a_i = \sum_j x_{ij}$ represent the loans to bank j by all other banks and bank i' loans to all other banks, respectively; and $\sum_i a_i = A$, $\sum_j l_j = L$, $A = L$.

Because we only know l_j and a_i, x_{ij} is unknown, i.e., X has N^2-2 N unknown elements. Therefore, the matrix X cannot be identified. To estimate x_{ij}, it is necessary to assume the distribution of the interbank bilateral lending data.

Suppose a and l are the realized values of the marginal distribution functions $f(a)$ and $f(l)$, respectively, x is the realized values of the joint distribution functions $f(a, l)$. If $f(a)$ and $f(l)$ are independent of each other, $x_{ij} = a_i l_j$. The assumption of independence means that the interbank loans are equally distributed to all other banks, thus forming a complete market network similar to Allen & Gale (2000).[7]

However, due to the assumption of independence, the diagonal elements of matrix X are non-zero, which implies that banks lend to themselves, which is not realistic. In order to resolve this issue, we construct a new matrix Xo, the elements of this matrix are as follows:

$$x_{ij}^0 = \begin{cases} 0, \forall i = j \\ a_i l_j, \text{otherwise} \end{cases} \tag{1}$$

Because X^0 violates the constraint $a_i = \sum_j x_{ij}$ and $l_j = \sum_i x_{ij}$, we look for another matrix X^*, which is as close to matrix X^0 as possible and satisfies the relevant constraints. In other words, we minimize the relative entropy between the two matrices to estimate the elements of X^* as follows:

$$\begin{aligned} & min \sum_{i=1}^{N} \sum_{j=1}^{N} x_{ij}^* ln \frac{x_{ij}^*}{x_{ij}^0} \\ & s.t. \sum_{j=1}^{N} x_{ij}^* = a_i x_{ij}^* \geqslant 0 \end{aligned} \tag{2}$$

The modified interbank bilateral lending relationship is represented by matrix X^* as follows:

[7] The independence assumption also excludes the existence of relational lending, which Upper & Worms (2004) consider to be an important feature of the interbank market.

$$
\begin{array}{c|ccccc|c}
 & & \sum_i & & & & \\
 & \multicolumn{5}{c|}{\begin{bmatrix} 0 & \cdots & x^*_{1j} & \cdots & x^*_{1N} \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ x^*_{i1} & 0 & x^*_{iN} & & \\ \vdots & \cdots & \ddots & \vdots & \\ x^*_{N1} & \cdots & x^*_{Nj} & \cdots & 0 \end{bmatrix}} & \begin{matrix} a_1 \\ \vdots \\ a_i \\ \vdots \\ a_N \end{matrix} \\
X^* = & & & & & & \\
\sum_j & l_1 & \cdots & l_j & \cdots & l_N &
\end{array}
$$

The optimization problem described by equation (2) involves setting, $x^0_{ij} = 0, x^*_{ij} = 0\ \forall$i = 1,2,…..n, and $ln(/0/0) \equiv 0$. Because the objective function is strictly concave, a unique solution to X* can be found using the RAS algorithm (Bacharach, 1965; Blien & Graef, 1997). Given matrix X^0, we set $\nu = 0$. The RAS algorithm is as follows:

In step 1, which involves matrix row scaling, ∀j = 1,2,3…..n, we define

$$\rho^{\nu}_i = \frac{a_i}{\sum_{\forall j x^0_{ij} \geqslant 0} x^{\nu}_{ij}}, \text{ to update } x^{\nu}_{ij} \leftarrow x^{\nu}_{ij}\rho^{\nu}_i, \tag{3}$$

In step 2, which involves matrix column scaling, $\forall j = 1, 2., ..., n$, we define

$$\delta^{\nu}_j = \frac{l_i}{\sum_{\forall i x^0_{ij} \geqslant 0} x^{\nu}_{ij}} \text{ to update } x^{\nu+1}_{ij} \leftarrow x^{\nu}_{ij}\delta^{\nu}_j, \tag{4}$$

In step 3, we replace $\nu \leftarrow \nu + 1$ back in step 1 and repeat steps 1 and 2 repeatedly until the matrix that satisfies the specified

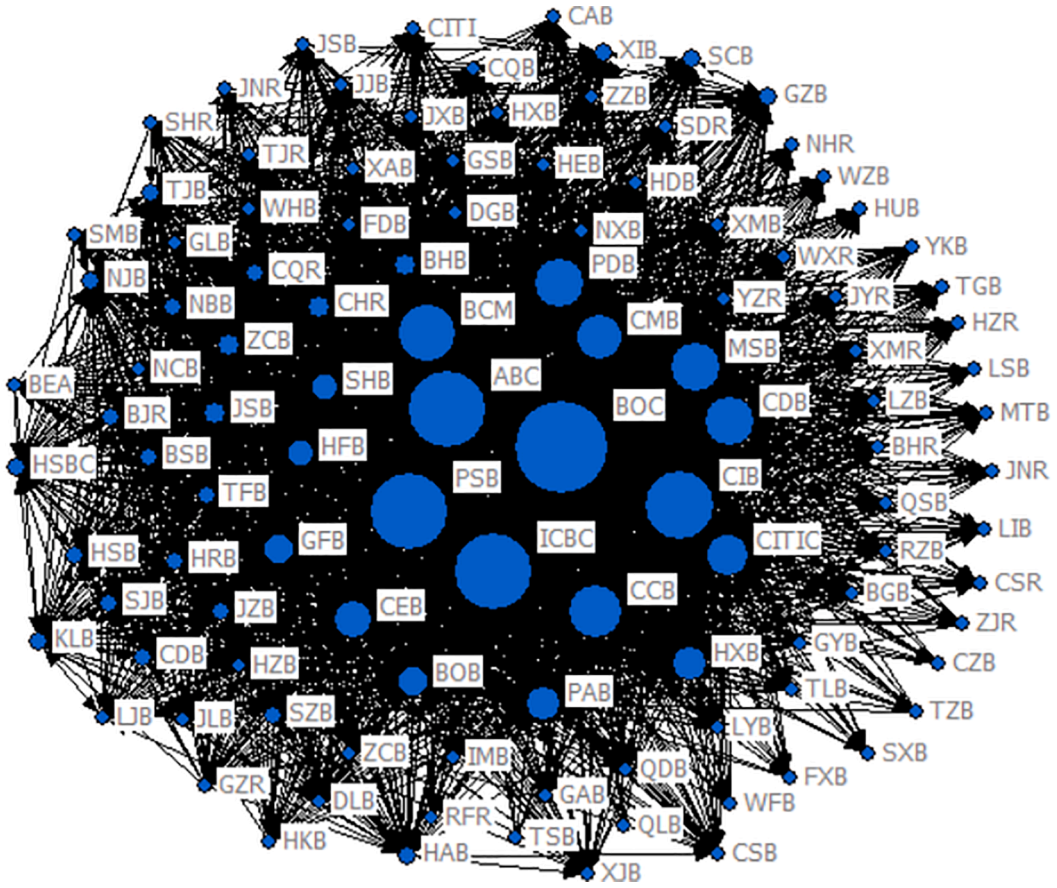


Fig. 1. China's banking network (2013).

conditions is generated. Once the RAS algorithm has been used to successfully estimate the interbank bilateral lending matrix, the risk contagion effect can be simulated by using the bank capital data.

3.2. Network

The bank bilateral lending matrix, $x_{ij} = a_i l_j$, is constructed based on the independence assumption of the maximum entropy principle. As the principal diagonal elements of the matrix are non-zero, to make the matrix economically meaningful, we normalize the matrix using the UCINET software. The diagonal elements of the resulting matrix are zero and the China's banking network is shown in Fig. 1.

The sample data shows very large variation among the banks. For example, the creditor's rights of Industrial and Commercial Bank of China (ICBC) to Bank of China (BOC) is US$24.43 billion, while Tangshan bank's creditor's rights to Dalian rural commercial bank is only US$0.07 billion. Accordingly, we set a threshold to further investigate the interbank connection and the network structure. First, we normalize the matrix and set the threshold value to 0.00001. Values less than 0.00001 indicate that the absence of any connection between banks (i.e., there is no bilateral lending). In Fig. 1, the connection between the nodes is weighted using the interbank lending data and this ratio is 1:7. The stronger the connection, the higher the interbank lending volume. The connection arrow points from the debtor bank to the creditor bank. The size of the node is proportional to the size of interbank lending, and this ratio is 1:5. The larger the node, the higher the interbank lending by the relevant bank. After setting the threshold value, the data shows that (i) the large banks are connected with all other banks, (ii) the small banks are generally connected with large banks, and (iii) the small banks are only loosely connected with each other, which forms an incomplete network structure (Allen & Gale, 2000) that is very similar to the German interbank market structure (Upper & Worms, 2004).

China's banking industry network, as shown in Fig. 1, has 4236 sides (the largest possible side is 12,210). Based on their connections and size, BOC, ICBC, ABC, PSB and Industrial Bank (CIB) are important nodes in the network. The interbank debt positions of these banks are the highest within the network. From the core/periphery index analysis, 12 banks (i.e., ICBC, CCB, ABC, BOC, China Development Bank (CDB), Bank of Communications (BCM), China Postal Savings Bank(PSB), China Merchants Bank (CMB), Shanghai Pudong Development Bank (PDB), CIB, Citic Bank (CITIC) and China Minsheng Bank (MSB)) are at the core layer of China's banking network. The rest of the banks are in the peripheral layer. These banks have few connections with other banks, and the connections are not strong, which is consistent with their relatively minor position in the network. Based on the core/peripheral indicators, if there is no connection between the peripheral and core banks then another incomplete network structure with higher concentration can be constructed (e.g., see in 't Veld and van Lelyveld, 2014; Fricke & lux, 2015; Chiu et al., 2020). In Section 4, using simulations, the contagion effect under the two incomplete network structures is considered and compared with the complete network structure.

4. Model simulation results

4.1. Sources of the contagion

4.1.1. Contagion caused by credit shocks

Suppose that x_{ji} represents the bank i's loan to bank j; λ is the default loss rate; and k_i is the capital of bank i. If $k_i < \lambda x_{ji}$, bank i goes bankrupt, which triggers the first round of contagion. We assume that an initial exogenous external shock causes the failure of bank j. If bank h extends loans x_{ih} and x_{jh}, respectively, to bank i and bank j, in the first round of contagion, bank j's failure would result in losses to bank $h$in the amount of λx_{jh}. However, if bank h did not go bankrupt, but bank i goes bankrupt due to the loss, bank h will also suffer losses in the amount of λx_{ih} in the second round of contagion. If $k_h < \lambda(x_{jh} + x_{ih})$, bank h will fail. Once bank h fails, it would turn to its creditor banks, which would trigger the third round of contagion. If banks continue to fail, the contagion will not stop until the loss arising from the shock is less than the capital of the affected bank. The factor affecting the contagion effect is the default loss rate λ, interbank debt size, and the bank capital k. The default loss rate λ is affected by the quality of bank assets, behaviour of banks in contagion, and the external economic conditions. In this paper, we assume that these conditions remain unchanged during the contagion.

4.1.2. Contagion caused by liquidity shocks

Under normal circumstances, banks are willing to roll over their interbank market debts that are close to maturity, but when banks are faced with significant liquidity constraints, the financing environment deteriorates rapidly, and it is difficult for the banks to roll over all short-term debt as it increases the liquidity risk. In the absence of new sources of capital, to maintain a reasonable balance sheet identity, banks sell their assets at a discount, which causes the loss of bank assets and can also have external effects.

Suppose that bank i has debt financing provided by bank h(the contagion mechanism presented here is valid even if debt financing comes from other banks). And bank h can only roll over $(1-\rho)$ of the financing under the stress scenario, there will be a financing gap of ρx_{ih}, this triggers bank i to sell assets at a discount of δ. As a result, the bank i sells assets worth $(1+\delta)\rho x_{ih}$ to make up for the financing gap, ρx_{ih}. In other words, the roll over risk contributes to a financing gap, which triggers fire sale of some assets and losses in the amount of $\delta\rho x_{ih}$. The losses are covered by bank i's capital. If $k_i < \delta\rho x_{ih}$ then bank i fails.

Haldane & May (2011) show that the contagion effect of a credit shock exhibits attenuation characteristics, while the contagion effect of a market liquidity shock shows amplification characteristics. The contagion effect of a financing liquidity shock does no exhibit attenuation characteristics, and both market liquidity and financing liquidity shocks produce external effects.

In the event of a credit and a liquidity shock at the same time, bank *i* will suffer from both assets and liabilities sides and hence the bank is more likely to fail. Therefore, in our simulations, we compare the contagion effect of a credit and a liquidity shock.

We have a sample of 111 bank and hence to examine the effect of a liquidity shock, we conduct 111 simulations, each of which involves a liquidity shock to one bank.

4.2. Contagion effect under a complete structure

In order to identify the sources of systemic risk in the banking industry, we start by simulating and comparing the contagion effect of a credit shock and a liquidity shock in 2013. The interbank lending matrix is estimated by Ras algorithm. Without setting a threshold value, except for the main diagonal element, all matrix elements are non-zero (i.e., all banks have claims and debts and thus the complete structure of the banking network can be identified). Next, we examine the contagion effect of a credit and a liquidity shock under the complete structure.

4.2.1. Contagion caused by a credit shock

When examining the impact of a credit shock, we assume that banks can extend the maturity of their interbank debt smoothly and hence the financing gap does not lead to sale of assets at a discount. A decline in bank assets arises from a default on bank debt. The default loss rate range is 0–1.

Under a complete network structure in 2013, the simulation results are presented in Table 2. Our simulation results show that, irrespective of the type of the capital used to counter the credit shock, even if the default loss rate reaches 1, the loss arising from the default by any bank is not strong enough to lead to failure of other banks. In real life, the default loss rate cannot be too high due to the external constraints arising from the legal protection mechanism. As a result, a credit shock is not expected to cause serious contagion.

4.2.2. Contagion caused by liquidity shocks

In recent years, more attention has been paid to the liquidity risk faced by the banking industry. China's financial market experienced a liquidity crunch in 2013 and 2016. When investigating the effect of a liquidity shock, we assume that the proportion of the interbank financing lies in the range 0–1 and the asset discount rate lies in the range is 0–2.[8] We then simulate the impact of the possible default effect (i.e., the non-rollover of the interbank debt) of each bank of the 111 banks in our sample on other banks in the network. The results presented in Table 3 exhibit some interesting characteristics.

First, when the core net tier 1 capital, which has the strongest ability to absorb losses, is used to counter the liquidity shock, the default by the BOC, ICBC, ABC, CCB, BCM, CIB and PDB can lead to the failure of other banks.

Under the extreme pressure scenario (i.e., ρ=1, δ=2), the BOC default can lead to failure of 21 banks, which will result is 40% loss of capital for the entire banking system. The contagion effect of this most important node in the banking network is shown in Figs. 2 and 3. Default on the part of ICBC can lead to failure of 9 banks. The resulting contagion effect, in terms of its adverse effect, is second only to that of the default by BOC. The default by CIB can lead to failure of 5 banks. As a joint-stock bank, the assets of CIB are much lower than that of the ICBC and BOC. However, the contagion effect is quite serious due to its heavy involvement in interbank business. In 2013, the interbank claims and debts ratio of the CIB reached 29.12% and 31.75%, respectively, which is significantly higher than other banks. Obviously, default by a large bank is more likely to lead to failure of other banks. This conclusion is confirmed by our simulations for different bank network structures, different shocks and contagion trends.

Based on 111 simulations involving a liquidity shock faced by each bank, we find that liquidity shocks lead to failure of 21 banks, including the PSB, 5 joint-stock banks, 2 rural commercial banks and 13 urban commercial banks. PSB and CDR failure is observed, respectively, 7 and 6 times in our simulations, whereas XMB failure occurs 4 times, and TFB and HRB fail 3 times, which are relatively more vulnerable nodes in the banking network.

Second, when net capital is used to counter the liquidity shock, the size of the contagion effect decreases. On the one hand, bank failures arising from the default decrease. For example, the default on the part of PDB no longer causes other banks to fail. The number of bank failures caused by BOC default decrease from 21 to 11. The number of bank failures caused by the ICBC default decrease from 9 to 5. On the other hand, the number of bank failures due to 111 liquidity shocks also decrease. Nine banks, including HXB, do not fail due to liquidity shocks. These results suggest that increase in the level of capital level can significantly enhance the stability of the banking system.

So far, we have considered the complete structure, we now consider the contagion effect under an incomplete structure.

[8] The data provided by China Monetary Network shows that, in December 2016, China's interbank lending transactions, i.e., one-day trading varieties (IB0001) accounted for 85.47% of all transactions. Pledged repurchase transactions, i.e., one-day trading varieties (R001) accounted for 78.42% of all transactions, when market funds were tight (such as June 2013 when funds were in short supply), the interest rates on the two types of transactions were high and hence the debt extension was adversely affected. Short-term wholesale financing markets (especially the ABCP) were essentially frozen after the collapse of Lehman Brothers in the U.S. subprime mortgage crisis, and it was very hard to roll over the interbank debt. Such a situation triggers asset sale and price declines. The discount rate of assets means that assets with a price of RMB 2 can only be sold at RMB 1. Moreover, the network analysis of Wells (2004) shows that banks face greater uncertainty while recovering creditor's rights and hence the default loss rate can be very high.

Table 2
Risk contagion effect of China's banking network under a credit shock in 2013.

Network Type	Capital Type	Banks causing failures (number of failures)	Failed banks (number of failures)
Complete Structure	Net core tier 1 capital	Zero	Zero
	Net capital	Zero	Zero
Incomplete structure I	Net core tier 1 capital	Zero	Zero
	Net capital	Zero	Zero
Incomplete structure II	Net core tier 1 capital	ABC (1), BOC (1), PSB (1)	HFB (3)
	Net capital	Zero	Zero

Note: The results presented in this Table are based on λ:0~1.

Table 3
Risk contagion in China's banking network under a liquidity shock in 2013.

Network Type	Capital Type	Banks causing failures (number of failures)	Failed banks (number of failures)
Complete Structure	Net core tier 1 capital	BOC (21), ICBC (9), CIB (5), BCM (3), CCB (2), ABC (2), PDB (1)	PSB(7), CDR(6), XMB(4), TFB(3), HRB(3), CIB(2), BHB(2), ZCB(2), JSB(2), HXB(1), GFB(1), TJB(1), GZB(1), KLB(1), XJB(1), JJB(1), CAB (1), GLB(1), WHB(1), HDB(1), JNR(1)
	Net capital	BOC (11), ICBC (5), CIB (4), ABC (2), BCM (2), CCB (1)	PSB (6), CDR (5), HRB (3), XMB (3), TFB (2), CIB (1), BHB (1), ZCB (1), JSB (1), GLB (1), HDB (1)
Incomplete structure I	Net core tier 1 capital	BOC (22), ICBC (9), BCM (5), CIB (5), CCB (2), ABC (2), PDB (1), CMB (1)	PSB(8), CDR(6), HRB(4), TFB(4), XMB(4), CIB(2), BHB(2), ZCB(2), JSB(2), HXB(1), GFB(1), TJB(1), HAB(1), GZB(1), KLB(1), XJB(1), JJB (1), CAB(1), GLB(1), WHB(1), HDB(1), JNR(1)
	Net capital	BOC (11), ICBC (5), CIB (4), BCM (2), ABC (2), CCB (1)	PSB (6), CDR (5), HRB (3), XMB (3), TFB (2), CIB (1), BHB (1), ZCB (1), JSB (1), GLB (1), HDB (1)
Incomplete structure II	Net core tier 1 capital	BOC (24), ICBC (14), BCM (5), CIB (5), ABC (3), CCB (2), PDB (2), CMB (1)	PSB(8), CDR(7), XMB(5), HRB(4), TFB(4), CIB(2), BHB(2), ZCB(2), TJB(2), XJB(2), JSB(2), GLB(2), WHB(2), HDB(2), MSB(1), HXB(1), GFB(1), BOB(1), HAB(1), GZB(1), KLB(1), JJB(1), CAB(1), JNR(1)
	Net capital	BOC (17), ICBC (6), BCM (5), CIB (5), ABC (2), CCB (2), PDB (1)	PSB(7), CDR(6), XMB(4), HRB(4), TFB(4), JSB(2), CIB(1), BHB(1), ZCB(1), GZB(1), KLB(1), XJB(1), JJB(1), CAB(1), GLB(1), WHB(1), HDB(1)

Note: The results presented in this Table are based on ρ:0~1, δ:0~2.

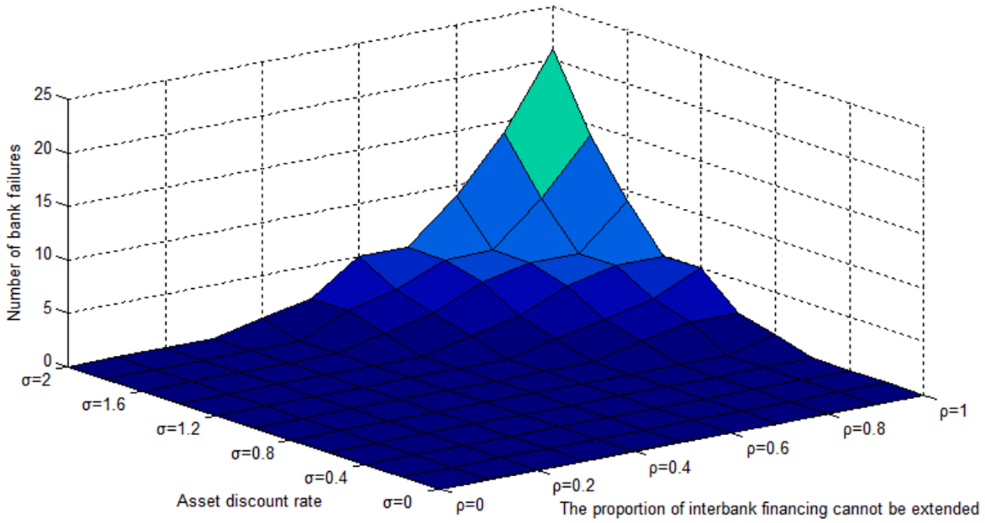


Fig. 2. Number of bank failures arising from the default by BOC.

4.3. Contagion effect under an incomplete structure

Owing to the factors such as the bank size, relationship lending, operating region, a bank usually borrows only from a few banks in the banking system. The relationship lending is an important feature of the interbank market (Upper & worms, 2004). Furfine (1999) shows that relationship lending prevails in the US interbank market. Affinito (2012) shows that the interbank relationships also persist over time in the Italian interbank market. Kobayashi and Taro (2018) propose a method to identify relationship lending and apply it to the Italian interbank networks. Using data from the e-MID market, they find that the proportion of relationship lending among all

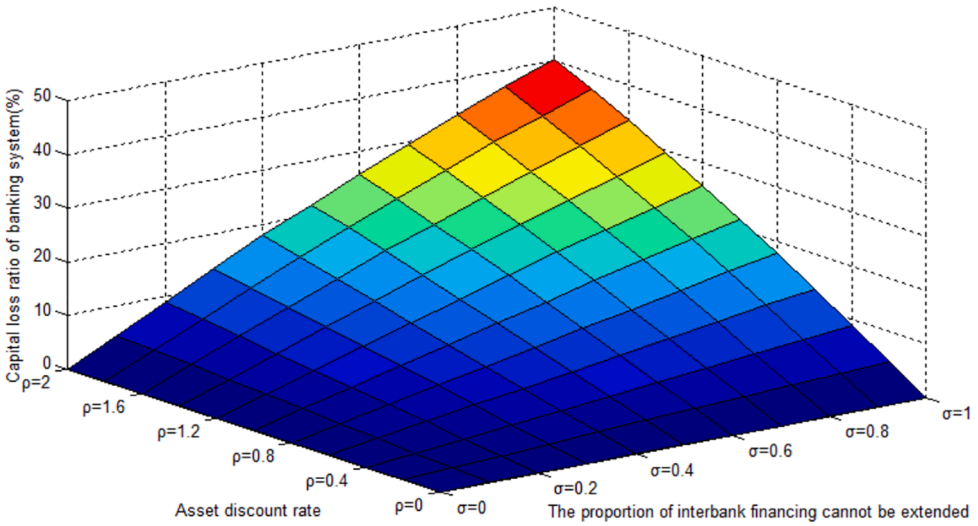


Fig. 3. Capital loss ratio of banking system arising from default by BOC.

bilateral trades has been quite stable. This conclusion is also supported by Temizsoy et al. (2015). Chiu et al. (2020) show that banks maintain long term relationships with their partners in the European interbank market. Based on the analysis of China banking network presented in Section 3, we assume that some banks in China's banking network do not borrow from each other, which allows us to construct two types of incomplete network structures.[9]

The first structure which involves a unilateral connection is based on a threshold as shown in Fig. 1. We use a threshold value of 0.00001 after normalization. There is no lending between the banks below the threshold but there is lending between banks above the threshold. Such a structure is known as an incomplete structure I because it involves a higher concentration of banks compared to the network without a threshold.

The second structure, which can be seen in Fig. 1 involves a situation where (i) the core layer banks borrow from each other (this is the case of a fully connected structure) and (ii) the core layer banks and the peripheral layer banks borrow from each other but there is no mutual lending between the peripheral banks. This network structure, which is highly centralized, is the incomplete structure II.[10]

We now consider the independence assumption along with the constraint discussed earlier. $x_{ij} = 0$, if $i = j$, or there is no mutual lending between *i* and*j*. We use the RAS algorithm to estimate the bank lending matrix and simulate the contagion effect. These results are also shown in Table 2. By comparing the simulation results for the two types of incomplete structure networks, we find that the contagion effect varies across the two incomplete structures.

4.3.1. Contagion caused by a credit shock

Under incomplete structure I, the contagion effect arising from a credit shock is similar to that of the complete structure. This continues to be the case even under the extreme pressure scenario (i.e., $\lambda = 1$). Irrespective of the capital (i.e., the net core tier 1 capital or net capital) used to counter the credit shock, default by none of the banks leads to failure of other banks (see Table 2). However, under incomplete structure II when the net core tier 1 capital is used to counter the credit shock, the default by ABC, BOC and PSB leads to the failure of HFB (HFB has faced a number of operational and management-related problems in the past few years and our simulation results reveal the fragility of the bank). Our simulation results also show that the impact on risk contagion varies across the bank structures.

4.3.2. Contagion caused by a liquidity shock

Under the extreme pressure scenario (i.e., ρ=1, δ=2), simulations were used to examine the contagion arising from a liquidity shock faced by each of the 111 banks. The contagion effects under different network structures are shown in Table 3.

First, we compare the contagion effects under incomplete network structure I and the complete structure. When using the net core tier 1 capital to counter the liquidity shock, in the incomplete network structure I, the default by CMB can lead to the failure of other banks. In the case of an incomplete structure, default by BOC leads to failure 22 banks (in the case of the complete structure, default by BOC lead to failure of 21 banks). Furthermore, in the case of the incomplete structure I, HAB fails due to the shock, and the number of bank failures due to default by PSB, TFB and HRB increases by 1. When net capital is used to counter the shock, the contagion effect does not change. In general, compared to the case of a complete network structure, the contagion effect under the incomplete network

[9] Gao & Pan (2012) use a similar assumption.

[10] Upper & Worms (2004) show that the German interbank market can be divided into two layers whereas Toivanen (2009) show that the Finnish interbank market can be divided into three layers.

structure I does not change much.

Second, as can be seen in Table 3, the contagion effect varies considerably across incomplete network structures I and II and the type of capital used to counter the shock. When net capital is used to counter the shock, the number of bank failures caused by BOC's default increase from 11 under incomplete network structure I to 17 under incomplete network structure II. However, this increase in smaller (from 22 to 24) when net core tier 1 capital is used. The use of net capital to counter the shock decreases the bank defaults under the incomplete structure II. As can be seen in Table 3, the default by ICBC under the incomplete structure leads failure of 14 banks, when net core tier 1 capital is used to counter the shock, but the bank failures decrease to only 6 when net capital is used. It is clear from Table 3 that the number of bank failures is higher under incomplete network structure II.

Third, the simulation results presented in Table 3 show that the default by BOC leads to most serious contagion effect across all 3 types of network structures and BOC is the most important node in the banking network in 2013. ICBC appears to be the second most important node in China's banking network. CIB default leads to most serious contagion effect out of the joint-stock banks. PSB and CDR appear to be relatively more vulnerable nodes in 2013 as these banks fail more frequently in our simulations.

Finally, our simulation results also show that the contagion effect, which varies across banking network structure, increases with banking network concentration.[11] It is worth noting that the theoretical analysis of Allen and Gale (2000) shows that under the complete banking network structure, depending on certain conditions, the interbank links can turn into a risk sharing channel. They show that under a different set of conditions, the interbank links can also turn into a risk contagion channel. However, they assume that banking network nodes are homogeneous. In real life banking networks, interbank creditor's rights and debt levels vary considerably and the nodes are heterogeneous and hence the empirical results can be very different. Moreover, as can be seen in our paper, the contagion characteristics of liquidity and credit shocks differ considerably.

4.4. Risk contagion to China's banking network from 2013 to 2016

4.4.1. Contagion caused by credit shocks

In this Section, we examine the trend in the contagion effect due to a credit shock over the 2013–2016 period. Under the complete network structure, the net core tier 1 capital was used to counter the shock. The simulation results presented in Table 4 reveal that default by any of the 111 banks considered in our study would not have resulted in failure of other banks in 2013 and 2015. If the default loss rate were 1 in 2014, the default by BOC would have led to the failure of CDB. If the default loss rate were 0.7 in 2016, the default by ICBC, ACB and CDB could have resulted in failure of SMB. If the default loss rate were 1, the default by BOC would have resulted in failure of SMB in 2016. Hence, CDB and SMB, respectively, would have been the most vulnerable nodes in 2014 and 2016 (see Table 4).

Analysis of the interbank claims and capital data reveals that the interbank claims of CDB in 2014 accounted for 766.80% of the net capital and 1020.22% of the net core tier 1 capital, ranking first among all banks. The interbank claims of SMB accounted for 1505.03% of the net capital and 1584.52% of the net core tier 1 capital in 2016, respectively, much higher than all other banks. For a very high interbank debt to capital ratio, once the debtor bank defaults, the loss is likely to exceed the level of capital, and the bank will easily fail. Thus, only a few banks would have failed due to a credit shock over the 2013–2016 period (and hence the contagion effect caused by the credit shock would not have been serious).

4.4.2. Contagion caused by liquidity shocks

In response to the potential risks associated with rapid development of the banking system, China's regulatory authorities took steps to strengthen the supervision of, among other things, interbank business from 2014 onwards. Increased supervision led to slowing down of the interbank business. In this section, using simulation results, we focus on changes to China's banking network risk contagion from 2013 to 2016. Some of the features can be observed in Table 5.

First, simulation results presented in Table 5 exhibit a downward trend in network risk contagion effect in China over the 2013–2016 period. The number of banks that would have failed due to the liquidity shock decreased significantly, which may be attributed to the slowdown in interbank business (i.e., deleveraging) in China. The liquidity shock caused by the default by 6 banks in 2015 could have led to the failure of CDB, which is the most vulnerable node in the network. Analysis of the CDB data shows that in 2015 the interbank debt accounted for 384.28% of the net capital (down from 623.41% in 2013) and 500.42% of the net core tier 1 capital (down from 690.27% in 2013). The corresponding proportions for BJR, which has fewer assets, in 2015 were 164.81% and 191.65%, respectively. Similarly, the liquidity shock caused by the default of 12 banks in 2016 could have led to the failure of SMB, which is the most vulnerable node in the network. The interbank liabilities of SMB account for 2527.03% of the net capital and 2660.5% of the net core tier 1 capital, respectively. The ratio of the SMB interbank liabilities to assets is as high as 38.92%. Obviously, the interbank debt to capital ratio is too high and hence the ability to counter the liquidity shocks is poor and hence SMB has become a vulnerable node in China's banking network.

Second, simulations across time periods show that default by a large bank can result in significant contagion in the Chinese banking network. The large banks include 5 state-owned banks, CDB and some joint-stock banks. However, the banks that would have failed due to the liquidity shock are mainly some joint-stock banks and a large number of small and medium-sized banks (mainly urban commercial banks). This situation highlights the asymmetric structure of China's interbank market financing to some extent, where the

[11] This result is consistent with Müller (2006). Müller finds that a centralized market is more prone to contagion.

Table 4
Risk contagion in China's banking network under a credit shock (2013–2016).

Year	Banks causing failures (number of failures)	Failed banks (number of failures)
2013	Zero	Zero
2014	BOC (1)	CDR (1)
2015	Zero	Zero
2016	ICBC (1), ABD (1), BOC (1), CDB (1)	SMB (4)

Note: The results presented in this Table are based on (i) the net core Tier 1 capital as a measure of capital, (ii) a complete network structure, and (iii) λ:0∼1.

Table 5
Risk contagion in China's banking network under a liquidity shock (2013–2016).

Year	Banks causing failures (number of failures)	Failed banks (number of failures)
2013	BOC (21), ICBC (9), CIB (5), BCM (3), CCB (2), ABC (2), PDB (1)	PSB(7), CDR(6), XMB(4), TFB(3), HRB(3), BHB(2), ZCB(2), JSB(2), HXB(1), GFB(1), TJB (1), GZB(1), KLB(1), XJB(1), JJB(1), CAB(1), GLB(1), WHB(1), HDB(1), JNR(1)
2014	CDB (12), ABC (4), ICBC (2), PSB (2)	HFB(4), BHB(4), KLB(2), ZCB(2), CIB(1), GFB(1), JSB(1), TJB(1), GZB(1), XIB(1), JZB(1), XMB(1)
2015	ICBC (1), CCB (1), ABC (1), BOC (1), CDB (1), CIB (1)	CDR (6)
2016	ICBC(1), CCB(1), ABC(1), BOC(1), CDB(1), BCM(1), CMB (1), CIB(1), CITIC(1), MSB(1), PDB(1), CEB(1)	SMB (12)

Note: The results presented in this Table are based on (i) the net core Tier 1 capital as a measure of capital, (ii) a complete network Structure, and (iii) setting ρ:0∼1, δ:0∼2.

large banks mainly play the role of liquidity providers, while small and medium-sized banks are mainly liquidity recipients. Compared to the large banks, small and medium-sized banks are at a disadvantage in terms of deposit absorption, ability to borrow from the central bank and alternative sources of capital. Consequently, small and medium-sized banks in China rely heavily on large banks for liquidity and are thus more vulnerable to liquidity shocks.[12] Moreover, unlike Pino and Sharma (2019),[13] we find that the contagion effects do not exhibit any regional characteristics. The banks that would have failed due to the liquidity shock include China Post Savings Bank, Industrial bank, Sichuan Tianfu bank, Zhuhai Huarun Bank and other regional banks. The absence of regional characteristics in our work could be attributed to the fact that all banks included in our sample are members of China Association of Interbank Market Dealers. The members of this association can directly interact with each other and hence the regional banks can also fully participate in the interbank market.

Third, the ability of the banks to cause contagion varies across time. For example, the CDB would not have caused any contagion in 2013, but this bank would have caused contagion over the 2014–2016 period. The default by CDB would have led to the collapse of 12 banks in 2014, which is the most important node in the network. In 2013, BOC was the most important node in the network. The contagion effect caused by CIB deserves attention. The contagion arising from the default by CIB is strong in 2013. The interbank claims of CIB were as high as US$175.67 trillion, which is not only significantly higher than CMB and other joint-stock banks with similar level of assets, but also higher than ICBC, CCB, CDB, and BCM. As a creditor bank, once CIB is unable to extend its interbank liabilities, the liquidity shock to the debtor banks is likely to be very serious. Under a stricter interbank business supervision regime, CIB's interbank claims declined sharply to US$16.56 trillion in 2016. However, its interbank liabilities show a significant upward trend over the 2013–2016 period, significantly higher than other banks with similar asset size.

The analysis presented in this Section suggests that the aim of banking sector stability can be achieved by continuous monitoring of the network and policies that can deal with ups and downs of the interbank business.

5. Concluding remarks and policy implications

Within the context of rapid growth of the interbank business and implementation of new approaches to bank capital management in China in 2013, this paper examines the risk contagion effect of the banking network. Using the interbank business data from 111 banks, we estimate the interbank bilateral lending matrix. We then simulate the risk contagion effect of China's banking network under credit and liquidity shocks. In order to examine the impact of contagion effect on China's banking network, we divided the banking network into a core and a peripheral layer. Banks located in the core layer include ICBC, CCB, ABC, BOC, CDB, BCM, PSB, CMB, PDB, CIB, CITIC and MSB. The rest of the banks are in the peripheral layer of China's banking network. Peripheral banks have relatively few connections with other banks and these banks hold a minor position in China's banking network.

[12] The turnover in the money market in 2013–2016 involves three types of net lenders. These are large commercial banks, policy banks and joint-stock banks. The top three types of net borrowers are rural financial institutions, urban commercial banks and securities companies (National Interbank Funding Center, 2020).

[13] Pino and Sharma (2019) investigated the risk contagion in the U.S. banking industry from 2001 to 2012. They argued that the banking crisis would only spread significantly in the local area, and only in banks with similar scale and banking regulatory characteristics.

While comparing the contagion effect of a credit and liquidity shock in 2013, we find that the contagion effect caused by a credit shock is not so obvious, even if the default loss rate was as high as 1, a credit shock cannot lead to failure of other banks. However, our simulations suggest that the contagion effect of a liquidity shock can be much more serious. Under the extreme pressure scenario (i.e., when $\rho = 1$ and $\delta = 2$), the liquidity shock caused by the default by 7 banks can lead to the failure of 21 banks. This suggests that liquidity shocks need to be taken much more seriously.

When the net core Tier 1 capital is replaced by net capital to absorb interbank default losses, the contagion effect significantly decreases, which implies that an increase in the level of capital can significantly enhance the stability of the banking system and the ability of the banks to withstand adverse shocks.

The simulation results show that the contagion effect arising from the shocks varies across banking network structure. An increase in concentration of the banking network gives rise to more serious contagion effect. However, the complex relationship between the banking network structure and risk contagion effect warrants further research.

Our simulations show that the level of contagion arising from a credit shock does not change much over the 2013–2016 period. On the other hand, the contagion due to the liquidity shock exhibits a significant downward trend over the same period. The downward effect could be attributed to improved supervision from 2014, which resulted in slowing down of the interbank business in China. Our simulations also identify the most important as well as the most vulnerable nodes in China's banking network.

We find that the contagion effect varies significantly across China's banking network structure. An increase in concentration of the network leads to stronger contagion effect. However, the complex nature of the relationship between banking network structure and risk contagion effect warrants further research.

The contagion effect arising from a credit shock does not change much over the 2013–2016 period. However, the effect of the liquidity shock shows a significant downward trend, which could be attributed to sticker supervision of the interbank business from 2014. The banks with very high interbank debt to capital and claim to capital ratios are relatively more vulnerable to both credit and liquidity shocks. In the case of a liquidity shock, the banks with contagion effect are mainly the large banks, while the banks which fail are mainly the joint-stock banks and a large number of small and medium-sized banks.

5.1. Policy implications

Simulation results presented in this paper highlight the need for a stable banking system. The Chinese banking system appears to be relatively more vulnerable to liquidity shocks. Some steps in this regard have already been taken by policy makers in China. For example, to improve the stability of the banking system, the China Banking Regulatory Commission (CBRC) has introduced some new liquidity risk monitoring indicators in recent years. For example, the liquidity coverage ratio (LCR) and liquidity ratio (LR) in September 2015 and the Net Stable Funding Ratio (NSFR), High-quality Liquidity Asset Adequacy Ratio (HQLAAR) and Liquidity Matching Rate (LMR) in December 2017. These indicators can potentially help improve the liquidity risk monitoring in China's banking system.

Simulation results also suggest that increase in the level of capital can improve the ability of the banks to withstand adverse shocks and hence banks must be encouraged to maintain adequate capital levels. Improved supervision of capital adequacy requirements is highly desirable, especially for small and medium-sized banks as these banks are more vulnerable to liquidity shocks. The overall capital ratio of China's banking institutions has remained at a high level in recent years (the core tier 1 capital adequacy ratios over the 2013–2016 period, respectively, were 9.95%, 10.56%, 10.91% and 10.8%, and the capital adequacy ratios over the same period were 12.19%, 13.18%, 13.45% and 13.3%, respectively).[14] However, small and medium-sized banks continue to face some operational problems, which led to a decline in capital adequacy ratios and in some cases the standards were not met. Accordingly, there is a need for continued monitoring of all banks.[15]

To reduce the severity of the potential contagion effect, steps need to be taken to enhance supervision of interbank business. Specifically, the level of interbank debt must be kept at a reasonable level. Small and medium-sized banks are relatively poorly equipped to withstand liquidity shocks and hence liquidity monitoring is highly desirable.

Finally, the banking network structure examined in this paper is relatively simple. The relationship between the banking network structure and contagion effect warrants further analysis.

CRediT authorship contribution statement

Bing Chen: Conceptualization, Methodology, Writing - original draft. **Li Li:** Conceptualization, Software. **Fei Peng:** Validation, Writing - review & editing. **Sajid Anwar:** Writing - review & editing.

[14] See CRBC (2020).

[15] China's banking regulatory authorities have stipulated that the core tier 1 capital adequacy ratio, tier 1 capital adequacy ratio and capital adequacy ratio of non-domestic systemically important banks as 7.5%, 8.5% and 10.5%, respectively. The quarterly financial statements released in September 2017 show that the three indicators for the Baoshang bank with problems in recent years were 7.38%, 7.38% and 9.52%, respectively. Similarly, at the end of 2018, the three indicators for Jinzhou bank were 6.07%, 7.43% and 9.12%, respectively, failing to meet the standards (CRBC, 2020).

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